Dhirubhai Ambani Institute of Information & Communication Technology
Mid Semester Test-1, 2nd Semester 2023- 24

Course Title	IT694 Computer Networks		**Max Marks**	15
Date	6 February 2024	**CLOSED BOOK**	**Time**	1 Hour

Be brief in your answers.

1. Write down the sequence of socket library functions invoked on the server side before a server is ready for communicating with the client using TCP. Give the important parameters and return values of these functions and write one sentence each about these functions. [3]

2. An HTTP client send the following GET message and the server's response follows. Answer the following questions, indicating **where in the message below** you find the answer. [4]

------------------------------------Client Message--

GET /cs453/index.html HTTP/1.1<cr><lf>Host: gaia.cs.umass.edu<cr><lf>
User-Agent: Mozilla/5.0 (Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax) <cr><lf>
*Accept:ext/xml, application/xml, application/xhtml+xml, text /html;q=0.9, text/plain;q=0.8, image/png, */*;q=0.5<cr><lf>*
Accept-Language: en-us, en;q=0.5<cr><lf>
Accept-Encoding: zip, deflate<cr><lf>
Accept-Charset: ISO-8859-1, utf-8;q=0.7,;q=0.7<cr><lf>*
Keep-Alive: 300<cr><lf>
Connection:keep-alive<cr><lf><cr><lf>

------------------------------------Server Response--

HTTP/1.1 200 OK<cr><lf>Date: Tue, 07 Mar 2008 12:39:45GMT<cr><lf>
Server: Apache/2.0.52 (Fedora) <cr><lf>
Last-Modified: Sat, 10 Dec2005 18:27:46 GMT<cr><lf>
ETag: "526c3-f22-a88a4c80"<cr><lf>
Accept-Ranges: bytes<cr><lf>
Content-Length: 3874<cr><lf>
Keep-Alive: timeout=max=100<cr><lf>
Connection:Keep-Alive<cr><lf>
Content-Type: text/html; charset=ISO-8859-1<cr><lf><cr><lf>
<!doctype html public "- //w3c//dtd html 4.0 transitional//en"><lf><html><lf>
<head><lf> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"><lf> <metaname="GENERATOR" content="Mozilla/4.79 [en] (Windows NT5.0; U) Netscape]"><lf>
<title>CMPSCI 453 / 591 /NTU-ST550ASpring 2005 homepage</title><lf></head><lf>
<much more document text following here (not shown)>

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a. What is the URL of the document requested by the browser?
b. What version of HTTP is the browser running?
c. Does the browser request a non-persistent or a persistent connection?
d. Was the server able to successfully find the document or not? What time was the document reply provided?
e. When was the document last modified?
f. How many bytes are there in the document being returned?

3. Consider the queuing delay in a router buffer. Let I denote traffic intensity; that is, I = La/R. **L** is the packet size, **R** is the transmission rate, and **a** denote the average rate at which packets arrive at the queue. Suppose that the queuing delay takes the form I / (1-I) * L / R for I < 1.
   a. Provide a formula for the total delay, that is, the queuing delay plus the transmission delay.
   b. Plot the total delay as a function of L /R. [4]

4. Suppose two hosts, A and B, are separated by 20,000 kilometres and are connected by a direct link of R = 2 Mbps. Suppose the propagation speed over the link is $2.5 * 10^8$ meters/sec. [4]
   a. Calculate the bandwidth-delay product $R*d_{prop}$.
   b. Consider sending a file of 800,000 bits from Host A to Host B. Suppose the file is sent continuously as one large message. What is the maximum number of bits that will be in the link at any given time?
   c. Provide a physical interpretation of the bandwidth-delay product.